Approved by Chair:

Signature

COURSE SECTION INFORMATION

Big Data Tools and Techniques II
Applied A.I. Solutions Development

Note: All academic inquiries will be replied to within three business days.

COURSE DESCRIPTION:

This course will explore in depth the current tools and frameworks used in the visualisation, streaming, and analytics of Big Data. Students will gain experience on the management of a big data project while applying current industry best practices. The focus of this course will be the creation of Big Data applications using tools such as Apache Spark and Apache Kafka.

COURSE OUTCOMES:

Upon successful completion of this course the students will have reliably demonstrated the ability to:

1. Compare and contrast the benefits and limitations of popular and widely used big data tools in big data ecosystem.
2. Build big data visualization dashboards using a variety of current industry tools.
3. Implement a big data analytics tool using Cassandra and Alteryx
4. Understanding of OLAP Cube and its consumption using PowerBI (create Dashboard)

LIST OF TEXTBOOKS AND OTHER TEACHING AIDS:

Reference:

Recommended Resources:

COURSE DELIVERY MODE:

Refer to the topical outline table for the delivery mode.

The instructional methods of this course are comprised of a combination of lectures, demonstrations, hands-on exercises and take-home assignments.

EVALUATION SYSTEM:

Assessment Tool:	Description:	Outcome(s) assessed:	EES assessed:	Date / Week:	% of Final Grade:
VM Setup	Individual	1, 2, 3, 4, 5	1,2,3,4,5,6,7,8,9,10,11	1-3	10
Final project	Group project	1, 2, 3, 4, 5	1,2,3,4,5,6,7,8,9,10,11		40
Finding valid datasources	Group project	1, 2, 3, 4, 5	1,2,3,4,5,6,7,8,9,10,11		10
InClass assignments	Group project	1, 2, 3, 4, 5	1,2,3,4,5,6,7,8,9,10,11		40

GRADING SYSTEM the passing grade for this course is: _D (50%)

A+	**90-100**	**4.0**	**B+**	**77-79**	**3.3**	**C+**	**67-69**	**2.3**	**D+**	**57-59**	**1.3**	**Below 50**	**F**	**0.0**
A	**86-89**	**4.0**	**B**	**73-76**	**3.0**	**C**	**63-66**	**2.0**	**D**	**50-56**	**1.0**			
A-	**80-85**	**3.7**	**B-**	**70-72**	**2.7**	**C-**	**60-62**	**1.7**						

Excerpt from the College Policy on Academic Dishonesty:
The *minimal* consequence for submitting a plagiarized, purchased, contracted, or in any manner inappropriately negotiated or falsified assignment, test, essay, project, or any evaluated material will be a grade of zero on that material.
To view George Brown College policies please go to www.georgebrown.ca/policies

TOPICAL OUTLINE

Week	Topic Task	Delivery mode	Outcome(s)	Content / Activities	Resources
1	1	In Person	Hands on practice / tutorials using Azure	- Introduction to Azure - Azure Data Lake - Azure Synapse Analytics	Resource material available on D2L
1	2	Online	Hands on practice / tutorials using Visual Studio	- Introduction to CUBE - Design and Deploy SSAS Tabular cube using Visual Studio - Consume Cube using Power BI	Resource material available on D2L
1	3	Online	Hands on practice / tutorials using Alteryx	- Introduction to Alteryx - Create Workflows using Alteryx common objects	Resource material available on D2L
2	1	In Person	Hands on practice / tutorials – knowledge so far	- Final project discussion - In class assignment	Resource material available on D2L

2	2	Online	Hands on practice / tutorials using Apache Spark	- Apachee Spark –notebook - Spark-Shell	Resource material available on D2L
2	3	In Person	Hands on practice / tutorials using Apache Spark	- Apachee Spark –notebook - Spark-Shell - Read from CSV/Mongo/MySQL	Resource material available on D2L
3	1	In Person	Hands on practice / tutorials using snowflakes dataplatform	- Snowflakes Data platform	Resource material available on D2L
3	2	Online	Hands on practice / tutorials – knowledge so far	- In class assignment	Resource material available on D2L
3	3	Online	Final Project	- Presentation for Final Project	Resource material available on D2L
	Please note: this schedule may change as resources and circumstances require. For information on withdrawing from this course without academic penalty, please refer to the College Academic Calendar: **http://www.georgebrown.ca/Admin/Registr/PSCal.aspx**				